MEZO BUILDATHON · WAVE 2 · TRACK 1 DEFI

DeVinci Blockchain

COIL.



Leveraged Bitcoin that survives redemptions.

Live on Mezo testnet · coil-mezo-blitz-bet.vercel.app

THE PROBLEM, READ ON MEZO MAINNET

On Mezo, borrowers lose positions to redemptions, not liquidations.

1,279

redemptions against **28** liquidations since launch

79.6

BTC taken from troves by redemption, for 5.94M MUSD

4

troves still open, for 19M MUSD in circulation

Source: MUSD TroveManager logs and state on Mezo mainnet · reprint with npm run stats

THE ANSWER

One transaction in. Protected while it lives. Always yours.

Coil opens a 1.5–4× MUSD position in one transaction instead of 25, then guards it against price drops, redemptions and Recovery Mode, without ever taking custody.

HOW A COIL WORKS

A flash swap from Mezo’s own pool, repaid by the trove it opens.

01

User signs

One EIP-712 message for the trove, plus the BTC deposit.

→

02

Flash swap

CoilRouter pulls extra BTC from the MUSD/BTC pool.

→

03

Trove opens

openTroveWithSignature, in the user’s name, with all the BTC.

→

04

Pool repaid

The minted MUSD settles the swap. Leftover goes back to the user.

Unwind and deleverage run the same path in reverse: closeTroveWithSignature / adjustTroveWithSignature.

WHY IT ONLY WORKS ON MEZO

Three Mezo primitives, used together.

1

Pool flash swaps

The MUSD/BTC pool sends first, calls hook(), then checks the invariant. Coil uses it as its flash lender and its curve as the quote.

2

MUSD signature ops

openTroveWithSignature and its siblings let a contract fund a trove for a borrower who signed. Coil never owns the trove.

3

Native BTC

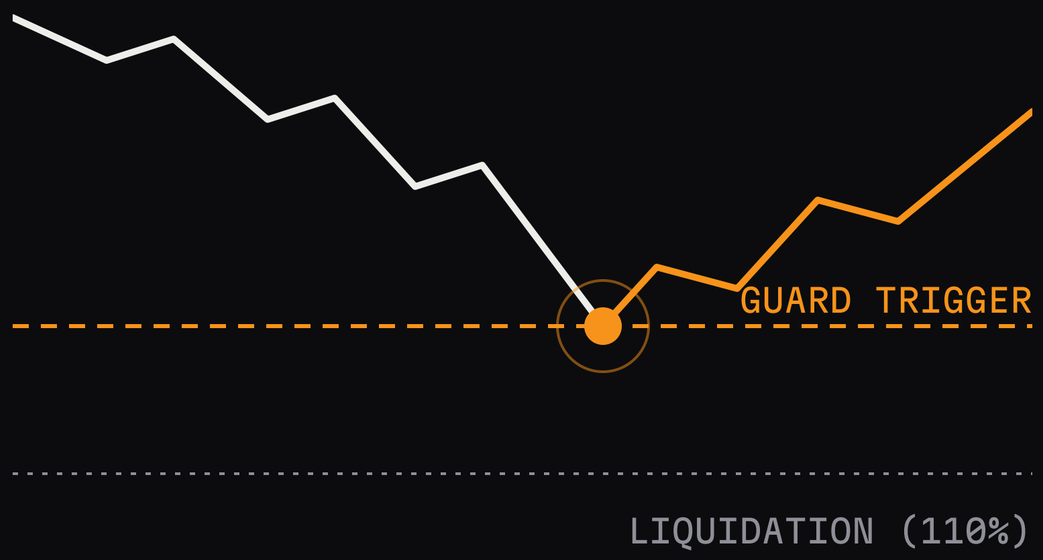
BTC is gas and an ERC-20 precompile over one balance, so BTC from the pool goes straight in as collateral. No wrapping.

GUARD

When BTC falls, the position defends itself.

The user signs a partial deleverage, a trigger and a minimum BTC out, once. When MUSD's price feed reaches the trigger, the always-on keeper fires it. Every swap is bounded by that same feed, so nobody can move the pool and sell the collateral cheap. Works on any MUSD trove.

Fired on-chain by the keeper · tx 0x76ce…2894



RESERVE, SHIELD, RECOIL · NEW IN WAVE 2

MUSD that earns while it waits, and repays when it must.

Reserve

The user parks MUSD in Mezo’s own sMUSD vault through Coil. It earns the savings rate. Only its owner can withdraw it.

Shield

A signed repayment from the Reserve. It fires at a price, or when less than X MUSD of debt stands ahead in the redemption queue, proved on-chain from SortedTroves.

Recovery Mode

When the system ratio drops under 150%, MUSD freezes collateral withdrawals, so a Guard can’t act. A Shield only repays, so it still can.

WAVE 1 REVIEW → WAVE 2

Every point of the review, answered.

THE REVIEW ASKED	WAVE 2 SHIPPED
Fix the High: bound fireGuard	PriceFeed bound on every swap + signed minBtcOut
Make it work in Recovery Mode	Shield: repay from the sMUSD Reserve
Run Guard on-chain with a keeper	Always-on keeper, polls every 15 s, fired on-chain
Wallets other than the deployer	Guard on any trove, Try-it page, outside wallets counted apart from team wallets
Verify CoilRouter	v2, v3, v4, lens and timelock source-verified on the Mezo explorer
Video ≤ 2 min, no forced intro	Under 2:00, the site opens straight

PROOF

Deployed, verified, fired on-chain, tested against the real Mezo.

107

unit, fuzz and invariant tests, 96.9% line coverage

9

fork tests on the real Mezo, testnet and mainnet

15 s

keeper reaction time, every run public on GitHub

`CoilRouter v4 0x17e056ef7e076Fbcb806dF9916552A26Da527F03 · verified · owner: 24 h timelock`
`Every transaction: coil-mezo-blitz-bet.vercel.app/#/proof`

THE PRODUCT

Everything a leveraged BTC holder needs, in one app.

FEATURE	WHAT IT DOES
Coil · Lever up · Unwind	Open, grow, shrink or close the trove in one transaction
Guard	Signed stop-loss, oracle-bounded, fired by the keeper
Reserve · Shield	sMUSD savings that repay before a redemption or in Recovery Mode
Recoil	Back to target leverage after BTC rises or a redemption
Risk cockpit	Liquidation odds, Monte Carlo, live redemption queue
Proof · Try it	Every transaction with a link; 2-minute tester onboarding

BUSINESS MODEL

Every Coil position grows Mezo’s fee base.

0.10%

Coil fee on each opening, capped at 0.5%. Live now: 0.05% of each coil’s debt goes to the MUSD/BTC gauge incentives.

sMUSD

MUSD interest flows to sMUSD savers through MUSD’s PCV. Every Coil trove pays it, and every Reserve holds sMUSD.

Keepers

0.30% of released BTC per Guard, 5 MUSD per Shield: a permissionless job.

Example: $5M of positions opened a month = $5,000 of Coil fees, and $5M of new MUSD debt paying interest to sMUSD.

ROADMAP

Built to stay after the Buildathon.

WAVE 1 · SHIPPED

- One-transaction coil, unwind, deleverage
- Guard
- Risk cockpit, radar

WAVE 2 · SHIPPED

- Oracle-bounded swaps
- Reserve, Shield, Recoil
- Gauge incentives, keeper
- v4 own tolerance, 24 h timelock

NEXT · Q1 2027

- Outside testers at scale
- External audit
- Mainnet once the pool is deep

LIVE ON MEZO TESTNET

Your Bitcoin, wound tight.



DeVinci Blockchain

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coil-mezo-blitz-bet.vercel.app · github.com/Wilivms/Coil · linkedin.com/in/wilivms